

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: cyber security

Practical 1: Implementing Substitution and Transposition Ciphers

Aim:

1.A Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques.

Code:

```
print("~~~~~T119 Dhairya Singh~~~~~")
print("Caesar Cipher")
def encrypt(text, shift):
    result = ""

    for char in text:
        if char.isalpha():
            if char.isupper():
                result += chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
            else:
                result += chr((ord(char) - ord('a') + shift) % 26 + ord('a'))
        else:
            result += char

    return result
```

```
def decrypt(cipher_text, shift):
```

```
    result = ""
```

```
    s
```

```
    for char in cipher_text:
```

```
        if char.isalpha():
```

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```
if char.isupper():
    result += chr((ord(char) - ord('A') - shift) % 26 + ord('A'))
else:
    result += chr((ord(char) - ord('a') - shift) % 26 + ord('a'))
else:
    result += char
return result

# Main Program
text = input("Enter the message: ")
shift = int(input("Enter the shift value: "))

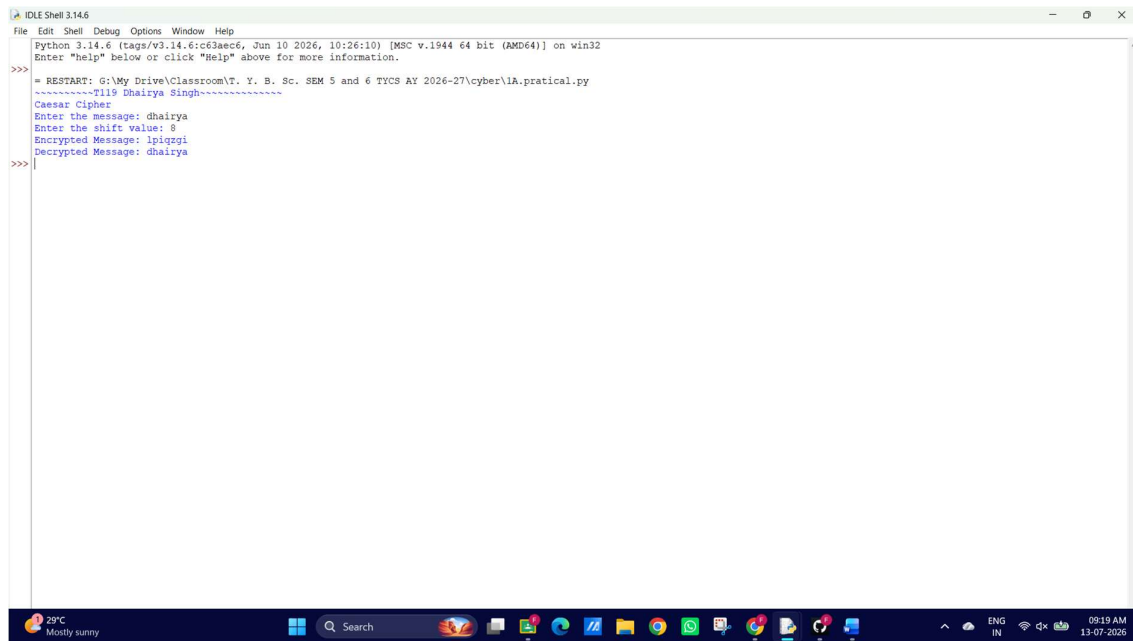
encrypted = encrypt(text, shift)
print("Encrypted Message:", encrypted)

decrypted = decrypt(encrypted, shift)
print("Decrypted Message:", decrypted)
```

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Output:



```
IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:63aeed6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: G:\My Drive\Classroom\T. Y. B. Sc. SEM 5 and 6 TYCS AY 2026-27\cyber\IA.practical.py
-----T119 Dhairya Singh-----
Caesar Cipher
Enter the message: dhairya
Enter the shift value: 8
Encrypted Message: lpiqqgi
Decrypted Message: dhairya
>>>
```

GUI:

```
import tkinter as tk
from tkinter import messagebox

# ----- Caesar Cipher Functions ----- #

def caesar_encrypt(text, shift):
    result = ""

    for ch in text:
        if ch.isalpha():
            if ch.isupper():
                result += chr((ord(ch) - ord('A') + shift) % 26 + ord('A'))
            else:
```

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```
        result += chr((ord(ch) - ord('a') + shift) % 26 + ord('a'))

    else:

        result += ch

    return result
```

```
def caesar_decrypt(text, shift):

    return caesar_encrypt(text, -shift)
```

```
# ----- Button Functions ----- #
```

```
def encrypt_text():

    try:

        text = txt_message.get()

        shift = int(txt_shift.get())

        encrypted = caesar_encrypt(text, shift)

        result_box.config(state="normal")

        result_box.delete(0, tk.END)

        result_box.insert(0, encrypted)

        result_box.config(state="readonly")
```

```
except ValueError:

    messagebox.showerror("Error", "Enter a valid shift value.")
```

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```
def decrypt_text():  
    try:  
        text = txt_message.get()  
        shift = int(txt_shift.get())  
  
        decrypted = caesar_decrypt(text, shift)  
  
        result_box.config(state="normal")  
        result_box.delete(0, tk.END)  
        result_box.insert(0, decrypted)  
        result_box.config(state="readonly")  
  
    except ValueError:  
        messagebox.showerror("Error", "Enter a valid shift value.")  
  
def clear_all():  
    txt_message.delete(0, tk.END)  
    txt_shift.delete(0, tk.END)  
  
    result_box.config(state="normal")  
    result_box.delete(0, tk.END)  
    result_box.config(state="readonly")
```

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----- GUI -----

```
root = tk.Tk()
```

```
root.title("Classical Substitution Cipher - Caesar Cipher")
```

```
root.geometry("500x320")
```

```
root.resizable(False, False)
```

```
title = tk.Label(
```

```
    root,
```

```
    text="Classical Substitution Cipher (Caesar Cipher)",
```

```
    font=("Arial", 15, "bold")
```

```
)
```

```
title.pack(pady=15)
```

```
# Message
```

```
tk.Label(root, text="Enter Message:", font=("Arial", 11)).pack()
```

```
txt_message = tk.Entry(root, width=50, font=("Arial", 11))
```

```
txt_message.pack(pady=5)
```

```
# Shift
```

```
tk.Label(root, text="Shift Value:", font=("Arial", 11)).pack()
```

```
txt_shift = tk.Entry(root, width=10, font=("Arial", 11))
```

```
txt_shift.pack(pady=5)
```

```
# Buttons
```

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```
frame = tk.Frame(root)
```

```
frame.pack(pady=15)
```

```
btn_encrypt = tk.Button(
```

```
    frame,
```

```
    text="Encrypt",
```

```
    width=12,
```

```
    bg="lightgreen",
```

```
    command=encrypt_text
```

```
)
```

```
btn_encrypt.grid(row=0, column=0, padx=10)
```

```
btn_decrypt = tk.Button(
```

```
    frame,
```

```
    text="Decrypt",
```

```
    width=12,
```

```
    bg="lightblue",
```

```
    command=decrypt_text
```

```
)
```

```
btn_decrypt.grid(row=0, column=1, padx=10)
```

```
btn_clear = tk.Button(
```

```
    frame,
```

```
    text="Clear",
```

```
    width=12,
```

```
    bg="tomato",
```

```
    command=clear_all
```

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)

```
btn_clear.grid(row=0, column=2, padx=10)
```

Result

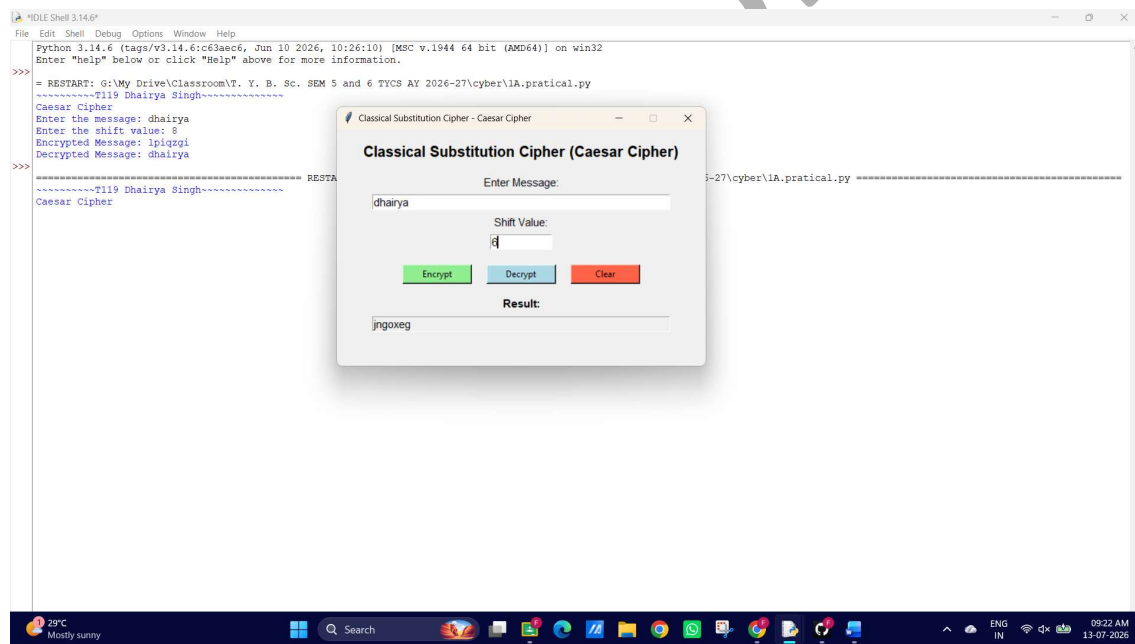
```
tk.Label(root, text="Result:", font=("Arial", 11, "bold")).pack()
```

```
result_box = tk.Entry(root, width=50, font=("Arial", 11), state="readonly")
```

```
result_box.pack(pady=5)
```

```
root.mainloop()
```

Output:

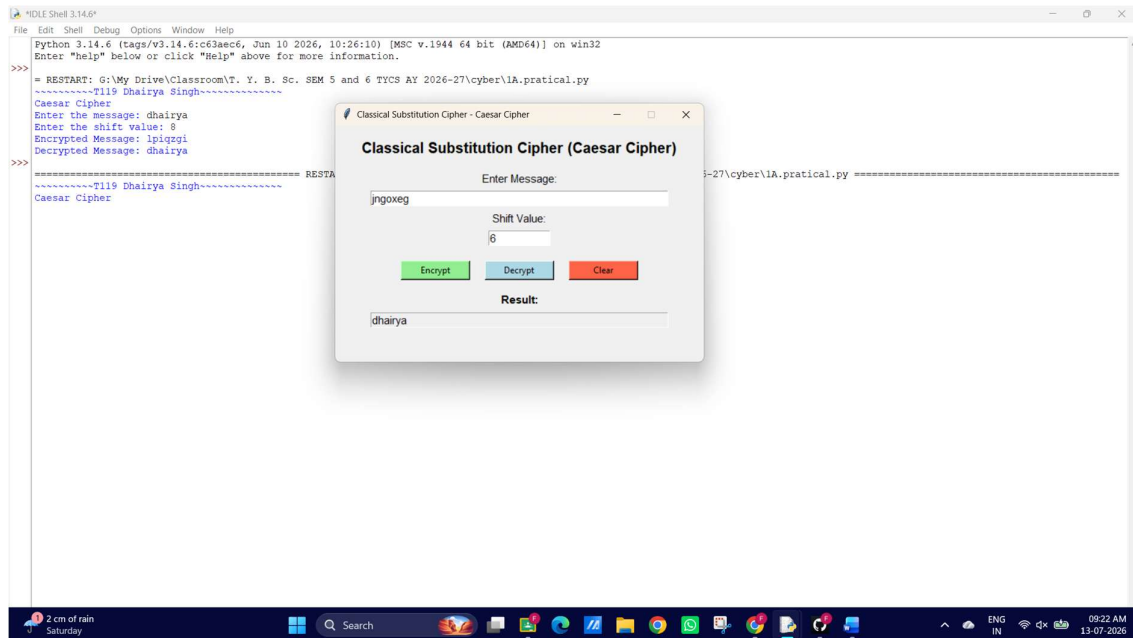


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1. B Design and implement algorithms to encrypt and decrypt messages using transposition techniques.

Code:

```
print("# ----- T119 Dhairya Singh ----- #")
print("Rail Fence Cipher (Transposition Technique)")
print()
# Encryption Function
def encrypt(text, key):
    rail = [['\n' for i in range(len(text))] for j in range(key)]

    direction_down = False

    row, col = 0, 0

    for char in text:
        if row == 0 or row == key - 1:
            direction_down = not direction_down
```

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```
direction_down = not direction_down
```

```
rail[row][col] = char
```

```
col += 1
```

```
if direction_down:
```

```
    row += 1
```

```
else:
```

```
    row -= 1
```

```
result = ""
```

```
for i in range(key):
```

```
    for j in range(len(text)):
```

```
        if rail[i][j] != '\n':
```

```
            result += rail[i][j]
```

```
return result
```

```
# Decryption Function
```

```
def decrypt(cipher, key):
```

```
    rail = [['\n' for i in range(len(cipher))] for j in range(key)]
```

```
    direction_down = None
```

```
    row, col = 0, 0
```

```
    for i in range(len(cipher)):
```

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```
if row == 0:  
    direction_down = True  
  
if row == key - 1:  
    direction_down = False
```

```
rail[row][col] = '*'  
col += 1
```

```
if direction_down:  
    row += 1  
else:  
    row -= 1
```

```
index = 0  
for i in range(key):  
    for j in range(len(cipher)):  
        if rail[i][j] == '*' and index < len(cipher):  
            rail[i][j] = cipher[index]  
            index += 1
```

```
result = ""
```

```
row, col = 0, 0
```

```
for i in range(len(cipher)):  
    if row == 0:  
        direction_down = True
```

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```
if row == key - 1:
```

```
    direction_down = False
```

```
result += rail[row][col]
```

```
col += 1
```

```
if direction_down:
```

```
    row += 1
```

```
else:
```

```
    row -= 1
```

```
return result
```

```
text = input("Enter the message: ")
```

```
key = int(input("Enter the key (number of rails): "))
```

```
encrypted = encrypt(text, key)
```

```
print("Encrypted Message:", encrypted)
```

```
decrypted = decrypt(encrypted, key)
```

```
print("Decrypted Message:", decrypted)
```

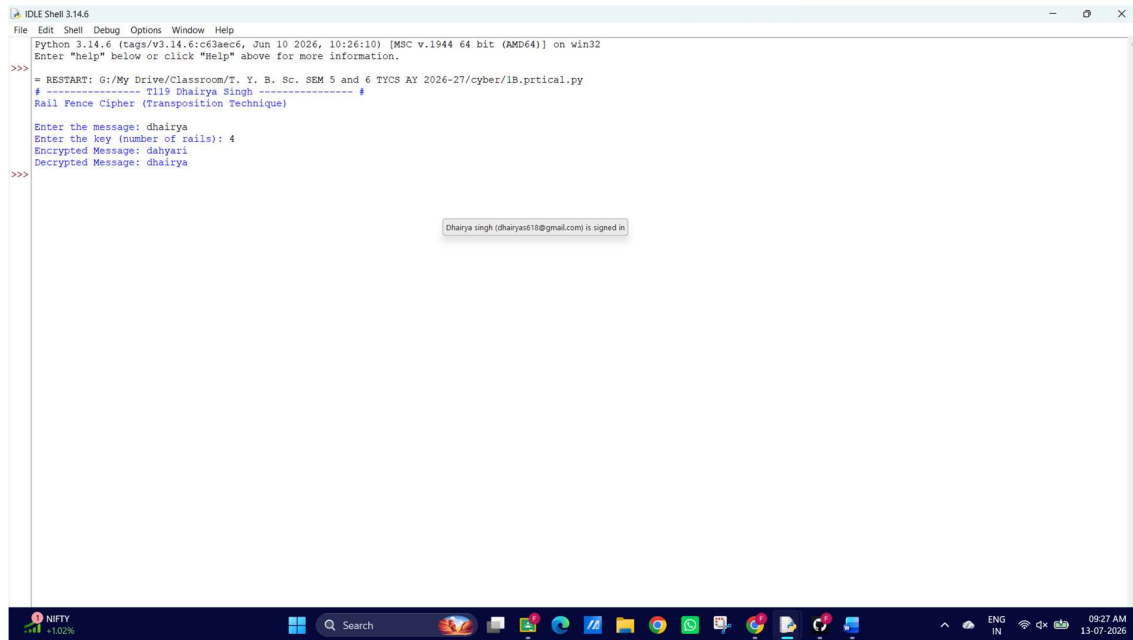
Output:

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```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: G:/My Drive/Classroom/T. Y. B. Sc. SEM 5 and 6 TYCS AY 2026-27/cyber/1B.ptical.py
# ----- T119 Dhairya Singh ----- #
Rail Fence Cipher (Transposition Technique)

Enter the message: dhairya
Enter the key (number of rails): 4
Encrypted Message: daivari
Decrypted Message: dhairya

>>>
```

GUI:

```
import tkinter as tk
from tkinter import messagebox

# ----- Encryption ----- #

def encrypt(text, key):
    if key <= 1:
        return text
```

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```
rail = [['\n' for _ in range(len(text))] for _ in range(key)]
```

```
row = 0
```

```
col = 0
```

```
direction_down = False
```

```
for ch in text:
```

```
    if row == 0 or row == key - 1:
```

```
        direction_down = not direction_down
```

```
    rail[row][col] = ch
```

```
    col += 1
```

```
    if direction_down:
```

```
        row += 1
```

```
    else:
```

```
        row -= 1
```

```
result = ""
```

```
for i in range(key):
```

```
    for j in range(len(text)):
```

```
        if rail[i][j] != '\n':
```

```
            result += rail[i][j]
```

```
return result
```

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```
# ----- Decryption ----- #
```

```
def decrypt(cipher, key):
```

```
    if key <= 1:
```

```
        return cipher
```

```
    rail = [['\n' for _ in range(len(cipher))] for _ in range(key)]
```

```
    row = 0
```

```
    col = 0
```

```
    direction_down = None
```

```
    # Mark zig-zag path
```

```
    for i in range(len(cipher)):
```

```
        if row == 0:
```

```
            direction_down = True
```

```
        if row == key - 1:
```

```
            direction_down = False
```

```
        rail[row][col] = '*'
```

```
        col += 1
```

```
    if direction_down:
```

```
        row += 1
```

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else:

row -= 1

Fill marked places

index = 0

for i in range(key):

for j in range(len(cipher)):

if rail[i][j] == '*' and index < len(cipher):

 rail[i][j] = cipher[index]

 index += 1

result = ""

row = 0

col = 0

for i in range(len(cipher)):

if row == 0:

 direction_down = True

if row == key - 1:

 direction_down = False

result += rail[row][col]

col += 1

if direction_down:

 row += 1

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else:

row -= 1

return result

def encrypt_text():

try:

text = message_entry.get()

key = int(key_entry.get())

if key < 2:

messagebox.showerror("Error", "Key must be greater than 1.")

return

result = encrypt(text, key)

result_entry.config(state="normal")

result_entry.delete(0, tk.END)

result_entry.insert(0, result)

result_entry.config(state="readonly")

except ValueError:

messagebox.showerror("Error", "Enter a valid key.")

def decrypt_text():

try:

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```
text = message_entry.get()

key = int(key_entry.get())

if key < 2:

    messagebox.showerror("Error", "Key must be greater than 1.")

    return

result = decrypt(text, key)

result_entry.config(state="normal")
result_entry.delete(0, tk.END)
result_entry.insert(0, result)
result_entry.config(state="readonly")

except ValueError:

    messagebox.showerror("Error", "Enter a valid key.")

def clear():

    message_entry.delete(0, tk.END)
    key_entry.delete(0, tk.END)

    result_entry.config(state="normal")
    result_entry.delete(0, tk.END)
    result_entry.config(state="readonly")
```

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```
root = tk.Tk()

root.title("Rail Fence Cipher - Transposition Technique")

root.geometry("550x330")

root.resizable(False, False)

title = tk.Label(
    root,
    text="Rail Fence Cipher (Transposition Technique)",
    font=("Arial", 16, "bold")
)
title.pack(pady=15)

tk.Label(root, text="Enter Message:", font=("Arial", 11)).pack()

message_entry = tk.Entry(root, width=50, font=("Arial", 11))
message_entry.pack(pady=5)

tk.Label(root, text="Enter Key (Rails):", font=("Arial", 11)).pack()

key_entry = tk.Entry(root, width=10, font=("Arial", 11))
key_entry.pack(pady=5)

frame = tk.Frame(root)
frame.pack(pady=15)

encrypt_btn = tk.Button(
```

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```
frame,

text="Encrypt",

width=12,

bg="lightgreen",

command=encrypt_text

)

encrypt_btn.grid(row=0, column=0, padx=10)


decrypt_btn = tk.Button(

    frame,

    text="Decrypt",

    width=12,

    bg="lightblue",

    command=decrypt_text

)

decrypt_btn.grid(row=0, column=1, padx=10)


clear_btn = tk.Button(

    frame,

    text="Clear",

    width=12,

    bg="tomato",

    command=clear )

clear_btn.grid(row=0, column=2, padx=10)

tk.Label(root, text="Result:", font=("Arial", 11, "bold")).pack()

result_entry = tk.Entry(root, width=50, font=("Arial", 11), state="readonly")

result_entry.pack(pady=5)
```

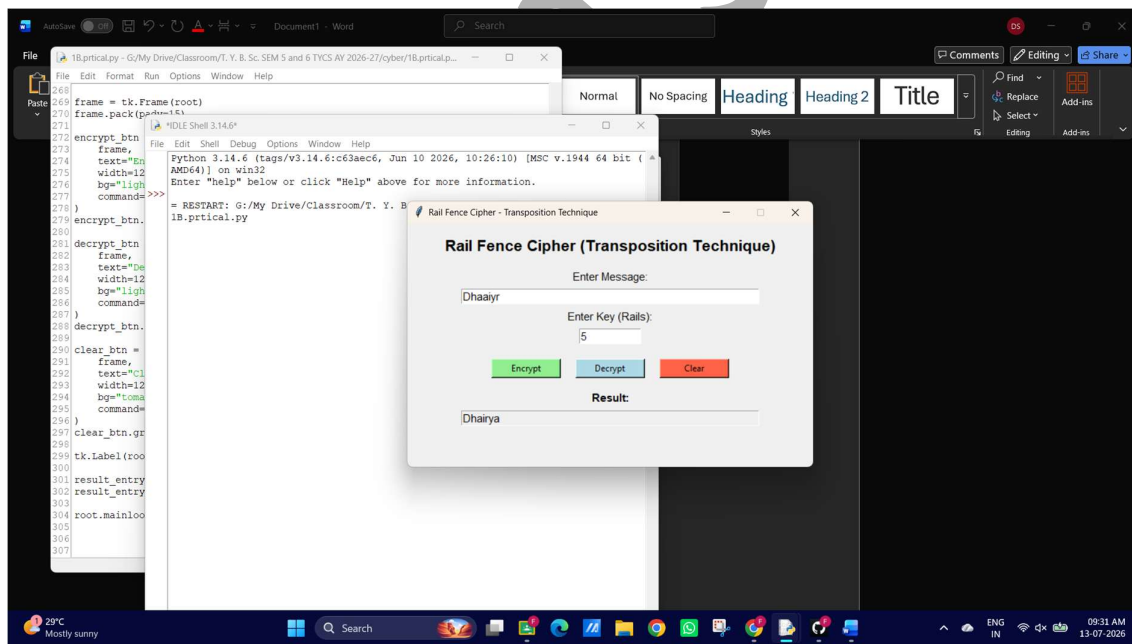
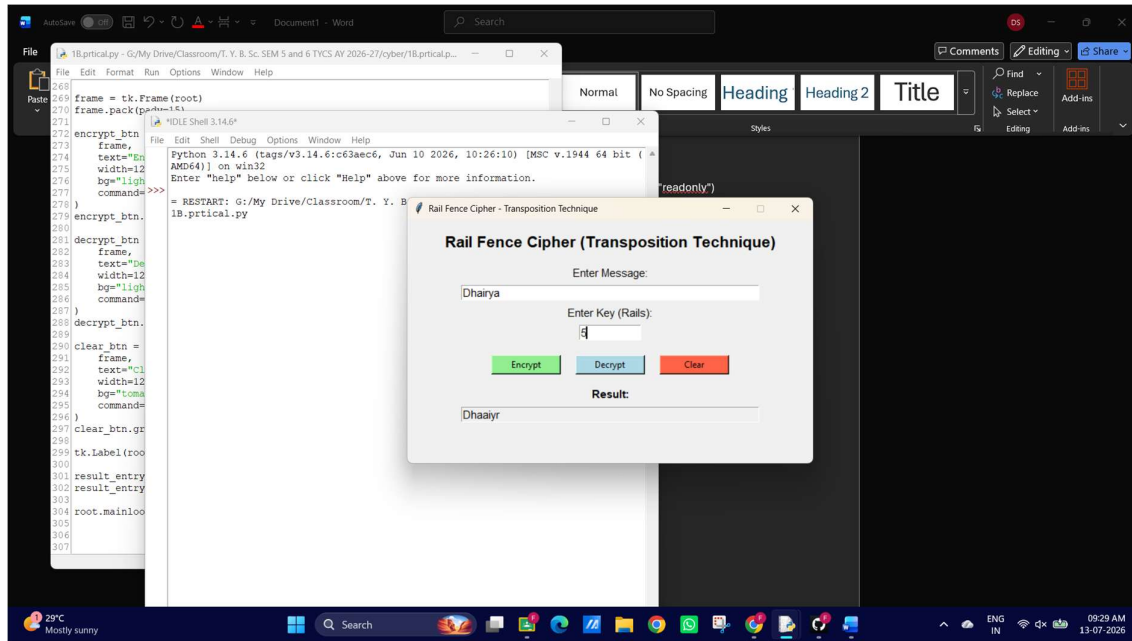
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root.mainloop()

Output:



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